

Name: _____

SOLVING LINEAR EQUATIONS, (ONE-STEP EQUATIONS)

Date: _____

LO: I can solve one-step linear equations using inverse operations.

What is a one-step equation?

A **one-step equation** needs only **ONE** inverse operation to solve, e.g. $x + 5 = 12$ needs only -5 .

Golden rule: whatever you do to one side, you must do to the OTHER side. This keeps the equation balanced.

Inverse operations

●	$+$ $-$	inverses
●	$\times$ $\div$	inverses
●	$x + 5 = 12$ $x = 7$	subtract 5
●	$3x = 12$ $x = 9$ (wrong: should $\div$)	$\div$ not $-$
●	Only do operation on one side	must do BOTH sides

Key idea

Use the INVERSE operation to undo what's done to x . Apply it to BOTH sides.

$$\boxed{x + 5} = 12 \quad \boxed{x = 7}$$

Subtract 5 from both sides

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - addition

Solve $x + 7 = 15$

Inverse of $+7$ is -7

$$x + 7 - 7 = 15 - 7$$

$$x = 8$$

$$\boxed{x = 8}$$

Check: $8 + 7 = 15$

Example 2 - subtraction

Solve $x - 9 = 4$

Inverse of -9 is $+9$

$$x - 9 + 9 = 4 + 9$$

$$x = 13$$

$$\boxed{x = 13}$$

Check: $13 - 9 = 4$

Example 3 - multiplication

Solve $5x = 35$

Inverse of $\times 5$ is $\div 5$

$$5x \div 5 = 35 \div 5$$

$$x = 7$$

$$\boxed{x = 7}$$

Check: $5 \times 7 = 35$

Example 4 - division

Solve $x \div 4 = 9$

Inverse of $\div 4$ is $\times 4$

$$x \div 4 \times 4 = 9 \times 4$$

$$x = 36$$

$$\boxed{x = 36}$$

Check: $36 \div 4 = 9$

YOUR TURN – PRACTICE (deliberate practice)

1. Add/subtract

Use the inverse operation.

- a) $x + 5 = 12$
- b) $x + 11 = 20$
- c) $x - 6 = 10$
- d) $x - 4 = 9$

2. Multiply/divide

Use the inverse operation.

- a) $3x = 18$
- b) $7x = 56$
- c) $x \div 5 = 6$
- d) $x \div 8 = 4$

3. With negatives

Be careful with sign changes.

- a) $x + 12 = 5$
- b) $x - 3 = -10$
- c) $-4x = 24$
- d) $x \div -3 = 5$

4. Word problems

Form an equation, then solve.

- a) I think of x . Add 9 to get 22. Find x .
- b) Multiply x by 4 to get 28. Find x .
- c) Divide x by 3 to get 12. Find x .
- d) Subtract 8 from x to get 0. Find x .

5. Mixed one-step

Identify the operation, then apply its inverse.

- a) $x + 17 = 25$
- b) $11x = 121$
- c) $x - 4.5 = 2.5$
- d) $x \div 6 = 9$
- e) $x + 2.4 = 5.1$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $x + 5 = 12$ means $x = 17$ ☐

Correct answer: _____

Reason: _____

b) $5x = 25$ means $x = 5$ ☐

Correct answer: _____

Reason: _____

c) Whatever you do to one side, do to the other ☐

Correct answer: _____

Reason: _____

d) Inverse of $\times 3$ is $\times 3$ ☐

Correct answer: _____

Reason: _____

e) $x \div 4 = 8$ means $x = 32$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Maya buys x cupcakes at **£1.20 each**, paying **£14.40** in total. Form a one-step equation and solve to find how many cupcakes she bought.

Expression: _____

Simplified: _____

b) A rectangle has perimeter **22 cm**. The length is **6 cm**. Form a one-step equation for the width w , and solve.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

